

ORENBURG, Russian Federation

WMO: 351210

Lat: 51.6925N

Lon: 55.0783E

Elev: 118

StdP: 99.92

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.5	(c) -26.2	(d) -32.9	(e) 0.2	(f) -29.4	(g) -29.4	(h) 0.3	(i) -26.0	(j) 11.7	(k) -9.4	(l) 10.4	(m) -9.6	(n) 1.7	(o) 80	(p) 0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.3	(c) 34.9	(d) 19.4	(e) 33.0	(f) 19.0	(g) 31.0	(h) 18.5	(i) 21.3	(j) 30.1	(k) 20.4	(l) 29.2	(m) 19.5	(n) 28.2	(o) 4.2	(p) 120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.5	24.5	17.3	12.5	23.5	16.3	11.8	22.8	62.6	30.1	59.1	29.0	56.2	27.9	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.3	(b) 9.1	(c) 8.1	(e) -33.8	(f) 37.5	(g) 3.4	(h) 1.9	(i) -36.2	(j) 38.9	(k) -38.1	(l) 40.0	(m) -40.0	(n) 41.0	(o) -42.4	(p) 42.4		
(a) 10.3	(b) 9.1	(c) 8.1	(e) -33.9	(f) 23.1	(g) 3.3	(h) 1.4	(i) -36.3	(j) 24.1	(k) -38.2	(l) 25.0	(m) -40.1	(n) 25.8	(o) -42.5	(p) 26.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-12.8	-11.9	-4.9	7.2	15.1	20.0	21.9	20.6	14.0	6.1	-2.5	-9.8
	DBStd	13.83	7.20	7.06	5.72	6.05	4.66	4.51	3.61	4.37	4.64	4.74	6.22	7.26
	HDD10.0	3059	707	613	461	123	11	1	0	0	16	139	377	612
	HDD18.3	5074	966	846	719	334	121	33	11	26	143	380	626	871
	CDD10.0	1365	0	0	0	39	170	302	369	331	136	18	0	0
	CDD18.3	338	0	0	0	1	22	84	121	98	13	0	0	0
	CDH23.3	4402	0	0	0	31	357	1078	1447	1262	224	3	0	0
	CDH26.7	1881	0	0	0	3	108	478	630	591	71	0	0	0
Wind	WSAvg	3.8	3.8	3.8	4.2	4.2	4.1	3.7	3.6	3.5	3.4	3.6	3.8	3.8
Precipitation	PrecAvg	365	28	21	21	24	29	40	42	27	29	36	32	32
	PrecMax	535	80	55	58	63	78	117	134	84	88	98	84	70
	PrecMin	217	6	1	0	0	3	0	0	0	2	2	5	6
	PrecStd	82	15	12	16	17	19	25	31	21	21	21	19	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.8	1.8	11.9	26.8	31.6	36.3	36.0	37.0	32.2	22.9	11.2	3.1
		MCWB	1.3	0.9	7.1	14.8	17.4	19.5	20.1	19.5	16.4	12.9	8.6	1.7
	2%	DB	0.2	0.9	5.2	23.9	29.7	34.1	34.0	34.9	28.4	19.9	8.4	1.3
		MCWB	-0.2	0.4	3.2	13.5	16.7	19.3	19.6	19.0	15.7	11.3	6.6	0.7
	5%	DB	-0.8	0.0	3.2	20.5	27.6	31.8	32.2	32.8	25.9	16.9	6.4	0.8
		MCWB	-1.2	-0.5	1.7	11.7	15.9	18.6	19.3	18.4	15.1	9.7	5.1	0.3
	10%	DB	-2.2	-1.8	2.0	17.5	25.0	29.3	30.3	30.0	23.0	13.9	4.8	-0.1
		MCWB	-2.7	-2.4	1.0	10.2	14.9	18.0	19.0	18.1	13.9	8.7	3.7	-0.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.1	1.1	7.4	15.7	19.2	22.2	22.8	22.7	19.3	14.3	9.6	2.1
		MCDB	1.3	1.3	11.2	25.2	28.0	31.7	31.0	32.1	29.3	20.0	10.6	2.5
	2%	WB	-0.1	0.3	3.4	14.0	17.7	20.7	21.6	20.6	16.7	12.0	7.1	0.9
		MCDB	0.1	0.6	4.9	22.5	27.0	30.2	30.0	29.7	25.4	17.7	8.1	1.2
	5%	WB	-1.2	-0.7	1.9	12.4	16.6	19.7	20.7	19.5	15.5	10.5	5.2	0.2
		MCDB	-0.9	-0.2	2.8	19.2	25.3	29.0	29.1	29.0	23.4	15.3	6.2	0.5
	10%	WB	-3.0	-2.3	1.0	10.7	15.6	18.7	19.8	18.6	14.5	9.2	3.7	-0.8
		MCDB	-2.5	-1.8	1.8	16.7	23.4	27.1	27.8	28.0	21.7	13.2	4.5	-0.3
Mean Daily Temperature Range	5% DB	MDBR	7.8	9.0	8.6	11.7	13.5	13.6	13.3	14.1	13.2	10.0	6.6	7.3
		MCDBR	5.3	5.9	7.9	16.2	17.0	16.8	15.9	17.2	17.8	15.6	7.8	5.3
		MCWBR	5.4	5.8	6.0	8.4	7.7	6.8	6.2	6.4	8.5	8.8	6.4	5.0
	5% WB	MCDBR	5.2	5.7	7.3	14.7	15.5	14.7	13.9	15.3	15.7	13.3	6.8	5.4
		MCWBR	5.4	5.8	5.8	8.2	7.8	6.5	6.0	6.6	8.1	8.2	6.1	5.3
Clear-Sky Solar Irradiance	taub	0.303	0.330	0.382	0.431	0.411	0.403	0.411	0.429	0.380	0.354	0.334	0.305	
	taud	2.067	2.010	1.960	2.078	2.238	2.322	2.303	2.228	2.328	2.316	2.216	2.082	
	Ebn at Noon	662	755	786	801	843	853	838	797	799	740	632	586	
	Edh at Noon	80	113	145	146	131	121	122	125	101	84	70	67	
All-Sky Solar Radiation	RadAvg	1.19	2.34	3.70	4.64	5.85	6.45	6.26	5.21	3.67	1.91	0.88	0.74	
	RadStd	0.15	0.17	0.42	0.43	0.48	0.57	0.43	0.39	0.39	0.30	0.14	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	+1.02	-0.61	-0.67	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+1.10	N/A	-0.57	N/A	N/A	N/A	N/A

Nomenclature: See separate page